Strengthened Results in the Cauchy-Sequence Completion of the Reals

A theorem-level comparison with `HITs/CauchyReals` and `HITs/CauchyRealsP`

Source audit of `cubical-master-reals`

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Abstract

The material improvement in the `Completion/CauchySequence/Reals` development is not that it proves ordinary completeness once again. It changes the logical strength of the input accepted by completeness and then consistently propagates that change through sequences, series, uniform convergence, power series, and integration. A Cauchy tail is chosen only at the radius currently being used, under propositional truncation. No global modulus is manufactured. Consequently, the new development proves the choice-free equivalence

$$\mathsf{C}_{\mathrm{rad}}(x) \simeq \mathsf{Lim}(x),$$

whereas the older sequential APIs start from a simultaneously chosen modulus. This yields genuinely stronger theorem interfaces: arbitrary-real rather than rational scaling of Cauchy series, sharp hypotheses such as $|x| < 1$ rather than a selected rational contraction, Mertens theorems without selected sum bounds, and uniform-limit/integration results without global modulus data. The report also records an important negative result: a suitably structured inverse from the regular completion to the new completion is equivalent to the missing sequential tail-choice principle. Finally, the older `HITs/CauchyReals` development still has substantially broader global calculus, logarithm, π, inverse trigonometry, circle, and complex-number coverage. Thus the new line is logically sharper in its sequential interfaces, not yet globally dominant in library coverage.

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1 Scope, names, and the comparison being made

There are three related, but not interchangeable, code lines in scope.

$\mathbb{R}_{\mathrm{C}}$ — **the HoTT-style higher inductive-inductive real.** `Cubical/HITs/CauchyReals`. Its constructors follow HoTT Book Definition 11.3.2: rational points, limits of $\mathbb{Q}_+$-indexed Cauchy approximations, and paths identifying points close at every radius.

$\mathbb{R}_{\mathrm{P}}$ — **the generic regular completion specialised to $\mathbb{Q}$.** `Cubical/HITs/CauchyRealsP`. This is the direct predecessor of most of the sequence/series/power-series code audited here. Its foundation re-exports `Relation/Premetric/Completion/Base`; its analysis API ordinarily exposes chosen sequential moduli.

$\mathbb{R}_{\mathrm{S}}$ — **the radiuswise Cauchy-sequence completion.** `Cubical/Relation/Premetric/Completion/CauchySequence/Reals`. Its limit constructor takes an $\mathbb{N}$-sequence and a separately truncated tail at each radius. Most direct old/new theorem pairs are therefore $\mathbb{R}_{\mathrm{P}}$ versus $\mathbb{R}_{\mathrm{S}}$.

The source contains an unconditional canonical nonexpanding map

$$c : \mathbb{R}_{\mathrm{P}} \longrightarrow \mathbb{R}_{\mathrm{S}}$$

which fixes rational points; companion modules prove preservation of distance, lattice, multiplication, and commutative-ring structure. No unconditional inverse is asserted. There is also no source theorem identifying the full `HITs/CauchyReals` development with $\mathbb{R}_{\mathrm{S}}$. Accordingly, this report does not silently transport old theorems across an unproved equivalence.

We use HoTT Book conventions: $\mathbb{Q}_+$ for positive rationals, $\|A\|$ for propositional truncation, dependent products and sums for $\prod$ and $\sum$, and $\simeq$ for equivalence. Write $x_m \sim_\varepsilon x_n$ for premetric closeness. This notation suppresses the implementation's two concrete closeness relations, which is immaterial to the logical comparison.

Classification used below. A strengthening weakens a premise, strengthens a conclusion, or removes data that cannot be recovered constructively. new coverage means a construction is new relative to $\mathbb{R}_{\mathrm{P}}$ but is not necessarily a stronger counterpart of a theorem already in $\mathbb{R}_{\mathrm{C}}$. A older side broader marks an area where the mature $\mathbb{R}_{\mathrm{C}}$ hierarchy remains broader.

2 The foundational material difference

2.1 The HoTT constructor and the new constructor

The HoTT Book and `HITs/CauchyReals` use Cauchy approximations. Schematically,

$$\mathsf{lim}_{\mathrm{C}} : \prod_{x:\mathbb{Q}_+\to\mathbb{R}_{\mathrm{C}}} \left(\prod_{\delta,\varepsilon:\mathbb{Q}_+} x_\delta \sim_{\delta+\varepsilon} x_\varepsilon \right) \longrightarrow \mathbb{R}_{\mathrm{C}}. \tag{1}$$

This is the free-completion idea of HoTT Book §11.3: limits and quotient paths are added simultaneously, avoiding a choice of representatives for a sequence in a quotient.

The new completion instead has a genuinely sequential constructor

$$\mathsf{lim}_{\mathrm{S}} : \prod_{x:\mathbb{N}\to\mathbb{R}_{\mathrm{S}}} \mathsf{C}_{\mathrm{rad}}(x) \longrightarrow \mathbb{R}_{\mathrm{S}}, \tag{2}$$

where

$$\mathsf{Tail}(x,\varepsilon) \;:\equiv\; \sum_{N:\mathbb{N}} \prod_{m,n:\mathbb{N}} (N \le m) \to (N \le n) \to (x_m \sim_\varepsilon x_n), \tag{3}$$

$$\mathsf{C}_{\mathrm{rad}}(x) \;:\equiv\; \prod_{\varepsilon:\mathbb{Q}_+} \|\mathsf{Tail}(x,\varepsilon)\|\,. \tag{4}$$

The one common cutoff needed at a particular ε exists merely; the definition contains no function selecting all cutoffs at once.

2.2 Three placements of truncation

The comparison module isolates three distinct propositions/data types:

$$\mathsf{C}_{\mathrm{ch}}(x) \;:\equiv\; \prod_{\varepsilon:\mathbb{Q}_+} \mathsf{Tail}(x,\varepsilon) \;\simeq\; \sum_{M:\mathbb{Q}_+\to\mathbb{N}} \prod_{\varepsilon:\mathbb{Q}_+} \text{tail-law}(M\varepsilon), \tag{5}$$

$$\mathsf{C}_{\mathrm{bd}}(x) \;:\equiv\; \|\mathsf{C}_{\mathrm{ch}}(x)\|\,, \tag{6}$$

$$\mathsf{C}_{\mathrm{rad}}(x) \;:\equiv\; \prod_{\varepsilon:\mathbb{Q}_+} \|\mathsf{Tail}(x,\varepsilon)\|\,. \tag{7}$$

There are constructive maps

$$\mathsf{C}_{\mathrm{ch}}(x) \longrightarrow \mathsf{C}_{\mathrm{bd}}(x) \longrightarrow \mathsf{C}_{\mathrm{rad}}(x),$$

but the missing reverse direction is exactly the passage

$$\left(\prod_{\varepsilon:\mathbb{Q}_+} \|A_\varepsilon\|\right) \longrightarrow \left\|\prod_{\varepsilon:\mathbb{Q}_+} A_\varepsilon\right\|. \tag{8}$$

That is a countable-choice principle over $\mathbb{Q}_+$, not a law of propositional truncation. The older $\mathbb{R}_{\mathrm{C}}$ sequential predicate `IsCauchySequence'` and $\mathbb{R}_{\mathrm{P}}$ predicate `isCauchySeq` inhabit the chosen side of this divide (their concrete packaging differs, but both expose simultaneous tail choices). The $\mathbb{R}_{\mathrm{S}}$ predicate inhabits the radiuswise side.

Finding 1 (Exact obstruction)**.** *Let* TC *be the real-specific sequential tail-choice statement that turns every radiuswise Cauchy sequence in the intermediate sequential presentation into one bundled sequential witness. The source proves*

$$\mathsf{StructuredInverse}(c) \simeq \mathsf{TC}. \tag{9}$$

Here "structured" means an equivalence $c : \mathbb{R}_{\mathrm{P}} \simeq \mathbb{R}_{\mathrm{S}}$ *whose inverse also supplies the bundled Cauchy witness demanded by the regular/sequential presentation. Propositional choice over* $\mathbb{Q}_+$ *implies* TC *and hence such an inverse. Bare invertibility of* c *is deliberately not claimed sufficient, because inverse values alone do not compute the missing tail witnesses.*

Equation (9) is the sharpest material comparison in the repository. It says where choice would enter and characterises exactly the extra structure it would buy. This directly echoes the warning in HoTT Book §11.3: quotient-first Cauchy completions need countable choice to lift a sequence of quotient elements to representatives. The new issue is not representative choice but tail choice; the same placement-of-truncation phenomenon is made explicit in the type.

3 The principal strengthened theorem

For a sequence $x : \mathbb{N} \to \mathbb{R}_{\mathrm{S}}$, define its type of limits by

$$\mathsf{isLim}(x, \ell) \;:\equiv\; \prod_{\varepsilon:\mathbb{Q}_+} \left\| \sum_{N:\mathbb{N}} \prod_{n:\mathbb{N}} (N \le n) \to (x_n \sim_\varepsilon \ell) \right\|, \tag{10}$$

$$\mathsf{Lim}(x) \;:\equiv\; \sum_{\ell:\mathbb{R}_{\mathrm{S}}} \mathsf{isLim}(x, \ell). \tag{11}$$

Separatedness makes $\mathsf{Lim}(x)$ a proposition. The constructor gives $\mathsf{C}_{\mathrm{rad}}(x) \to \mathsf{Lim}(x)$; the triangle inequality gives $\mathsf{Lim}(x) \to \mathsf{C}_{\mathrm{rad}}(x)$ without choosing tails at all radii. Therefore:

$$\boxed{\mathsf{C}_{\mathrm{rad}}(x) \;\simeq\; \mathsf{Lim}(x).} \tag{12}$$

In source this is `isCauchySequence≃sequenceLimit`. The older direct counterpart `cauchySequenceLimit` proves only

$$\mathsf{C}_{\mathrm{ch}}(x) \longrightarrow \mathsf{Lim}(x).$$

Thus equation (12) is stronger in both directions: its forward map accepts less structured Cauchy evidence, and its reverse map establishes a full criterion for convergence. It does *not* imply $\mathsf{C}_{\mathrm{rad}}(x) \to \mathsf{C}_{\mathrm{bd}}(x)$; completeness returns a real limit, not a global modulus.

Two important corollaries are then obtained at the correct strength:

$$\begin{aligned} &f \text{ continuous},\ \mathsf{C}_{\mathrm{rad}}(x) \quad\Longrightarrow\quad \mathsf{C}_{\mathrm{rad}}(f \circ x), \\ &f \text{ nonexpanding},\ \mathsf{isLim}(x, \ell) \quad\Longrightarrow\quad \mathsf{isLim}(f \circ x, f\ell). \end{aligned}$$

The first implication uses completeness of $\mathbb{R}_{\mathrm{S}}$: obtain the unique limit, apply sequential continuity, then use convergence $\Rightarrow$ radiuswise Cauchy. It would be false for an arbitrary incomplete target space, so the proof is not merely a repackaged Lipschitz lemma.

4 Theorem-by-theorem strength ledger

Area / class	**Earlier statement or burden**	**Strengthened $\mathbb{R}_{\mathrm{S}}$ result and why it matters**
Cauchy predicate strengthening	$\mathsf{C}_{\mathrm{ch}}(x)$: a cutoff at every radius is simultaneously selected.	$\mathsf{C}_{\mathrm{rad}}(x) = \prod_\varepsilon \lVert\mathsf{Tail}(x, \varepsilon)\rVert$. This is implied by the old predicate; the converse is precisely tail choice and is not assumed.
Completeness strengthening	`cauchySequenceLimit`: $\mathsf{C}_{\mathrm{ch}}(x) \to \mathsf{Lim}(x)$.	`isCauchySequence≃sequenceLimit`: $\mathsf{C}_{\mathrm{rad}}(x) \simeq \mathsf{Lim}(x)$. This is the central stronger result, not merely a new constructor.
Eventually equal and ordered strengthening	The established predicates store $\sum_N \prod_{n \ge N}(\cdots)$ and expose N.	`EventuallyEqualSequence`, `EventuallyLeSequence`, and `EventuallyLtSequence` truncate the cutoff. Eventual $\le$ and even eventual $<$ pass to $\le$ of limits without choosing a witness.

Area / class	**Earlier statement or burden**	**Strengthened $\mathbb{R}_S$ result and why it matters**
Asymptotic closeness strengthening	A selected cutoff is returned at each radius and clients often retain the witness.	`EventuallyClose` follows the same radiuswise truncation discipline and preserves Cauchyness and limits by proposition-valued elimination.
Series criterion strengthening	$\mathsf{CauchySeries}(a) \to \mathsf{Summable}(a)$, with the former built from a chosen modulus of partial sums.	`isCauchySeries`$\simeq$`summable` gives an equivalence at the radiuswise strength: $$\mathsf{C}_{\mathrm{rad}}\left(n \mapsto \sum_{k<n} a_k\right) \simeq \mathsf{Sum}(a).$$ No global series modulus is recovered or required.
Scalar closure strengthening	`rationalScaleSeriesIsCauchy` and `rationalScaleSeriesIsAbsolutelyCauchy` only scale by $q : \mathbb{Q}$.	`scaleSeriesIsCauchy` and `scaleSeriesIsAbsolutelyCauchy` accept every $c : \mathbb{R}_S$. Native completeness plus continuity of multiplication removes the rational-only restriction.
Majorants strengthening	From $\lvert a_n\rvert \le b_n$, old power-series theorems additionally request $0 \le b_n$ for all n.	`coefficientMajorantIsNonnegative` derives $$0 \le \lvert a_n\rvert \le b_n,$$ so all coefficient-majorant convergence theorems drop the redundant nonnegativity premise.
Absolute bounds strengthening	Mertens and iterated-product clients select explicit $A : \mathbb{Q}_+$ bounding an absolute sum.	`mereAbsoluteCauchySeriesSumBound` proves only $$\left\lVert \sum_{A:\mathbb{Q}_+} \mathsf{AbsSumBound}(A, a) \right\rVert.$$ Clients open such a bound only inside a proposition. The useful conclusion survives while the arbitrary bound is removed from the public theorem.
Mertens product strengthening	Old absolute Mertens statements take $A, B : \mathbb{Q}_+$ and proofs that the two absolute sums are bounded by them; separate theorems establish ordinary or absolute Cauchyness.	For the Cauchy product $(a *_{\mathrm{C}} b)_n = \sum_{k\le n} a_k b_{n-k}$: $$\mathsf{AbsC}(a) \to \mathsf{AbsC}(b) \to \mathsf{AbsC}(a *_{\mathrm{C}} b),$$ $$\sum(a *_{\mathrm{C}} b) = (\sum a)(\sum b).$$ Theorems `absoluteMertensCauchyProductIsAbsolutelyCauchy` and `absoluteMertensSeriesSum-cauchyProduct` need no selected sum bounds.

Area / class	**Earlier statement or burden**	**Strengthened $\mathbb{R}_S$ result and why it matters**
Geometric series strengthening	Callers provide $\mathsf{RatContr}(x) \coloneqq \sum_{q:\mathbb{Q}_+} (\lvert x\rvert < q) \times (q < 1)$.	`geometricSeriesIsAbsolutelyCauchyOfAbs<1` and its summability/value variants assume only the sharp open condition $\lvert x\rvert < 1$. Density yields $\lVert\mathsf{RatContr}(x)\rVert$, which is opened only into the propositional conclusion.
Bounded power series strengthening	With $\lvert a_n\rvert \le C$, convergence at x requires a selected $\mathsf{RatContr}(x)$.	`powerSeriesIsAbsolutelyCauchyOfAbs<1`, `powerSeriesIsAbsolutelySummableOfAbs<1`, and `powerSeriesIsSummableOfAbs<1` require only $\lvert x\rvert < 1$. Product and interior convergence layers reuse the same sharp interface rather than leaking an intermediate rational radius.
Uniform series strengthening	A whole uniform modulus $M : \mathbb{Q}_+ \to \mathbb{N}$ is part of `UniformlyCauchySeries`.	The new definition is $\mathsf{UCauchy}(S) \coloneqq \prod_{\varepsilon:\mathbb{Q}_+} \lVert\mathsf{UniformTail}(S,\varepsilon)\rVert$. Here $\mathsf{UniformTail}$ contains one cutoff common to all parameters. Thus the definition does not collapse to pointwise convergence. It yields uniform convergence to the canonical sum and continuity of that sum with no chosen global modulus.
Uniform continuity strengthening	Conventional APIs package a modulus choosing an input radius for every requested output radius.	`IsRadiuswiseUniformlyContinuous` uses $\mathsf{RU}(f) \coloneqq \prod_{\varepsilon:\mathbb{Q}_+} \lVert\mathsf{ModulusAt}(f,\varepsilon)\rVert$. Here $\mathsf{ModulusAt}$ contains an adequate input radius and its preservation law. Conventional uniform continuity implies $\mathsf{RU}(f)$; no converse is exported because it would again commute product with truncation.

Area / class	Earlier statement or burden	Strengthened $\mathbb{R}_S$ result and why it matters
Uniform limits and integration strengthening / new coverage	The older $\mathbb{R}_P$ series hierarchy has no Riemann-integration layer; the older $\mathbb{R}_C$ layer uses stronger uniform-continuity/modulus interfaces.	An arbitrary uniformly convergent sequence of radiuswise uniformly continuous functions has a radiuswise uniformly continuous limit. The completed dyadic integral preserves both global and interval-local uniform limits: $$f_n \to f \text{ uniformly on } [a,b] \implies \int_a^b f_n \to \int_a^b f.$$ Completing the sequence of integrals independently is proved equal to integrating the limit.
Transcendental construction new coverage	$\mathbb{R}_P$ stops at generic power-series machinery. $\mathbb{R}_C$ already has a different, mature transcendental hierarchy.	$\mathbb{R}_S$ now constructs power-series exp, sin, cos, hyperbolic functions, a partial tangent layer, local arctangent/reciprocal results, and $-\log(1-x)$ on $\lvert x\rvert < 1$, with addition, derivative, order, and identity theorems. This is substantial new coverage relative to $\mathbb{R}_P$, but not by itself a stronger theorem than the corresponding mature $\mathbb{R}_C$ results.

5 Four representative proof improvements

5.1 Sharp open hypotheses without selected rational witnesses

The geometric-series change is the cleanest recurring proof pattern. The old implementation already proves convergence from $\mathsf{RatContr}(x)$. Rational density proves only

$$|x| < 1 \quad \longrightarrow \quad \|\mathsf{RatContr}(x)\|,$$

which is exactly what constructive density should provide. In $\mathbb{R}_S$, absolute Cauchyness is a proposition, so truncation elimination gives

$$|x| < 1 \quad \longrightarrow \quad \mathsf{AbsC}(n \mapsto x^n)$$

without selecting q. The same proof architecture supplies the sharp hypotheses for bounded power series and their products. The mathematical condition visible to a client is now the standard one; the rational q remains an internal estimate.

This is materially stronger than adding a convenience lemma only if the target is genuinely propositional. The report therefore does not claim a construction of a canonical contraction radius. Such a construction is neither present nor needed.

5.2 Mertens without bound plumbing

For an absolutely Cauchy series a, Archimedean boundedness gives

$$\mathsf{AbsC}(a) \longrightarrow \left\| \sum_{A:\mathbb{Q}_+} \mathsf{AbsSumBound}(A, a) \right\|.$$

The old Mertens interface made A and its proof caller-visible. The new proof eliminates mere bounds for a and b directly into the proposition that rectangular partial sums approximate triangular Cauchy-product sums. It then derives, in order,

$$\begin{aligned} \mathsf{AbsC}(a), \mathsf{AbsC}(b) &\Longrightarrow \mathsf{RectangularApproximation}(a, b), \\ &\Longrightarrow \mathsf{Cauchy}(a *_{\mathrm{C}} b), \\ &\Longrightarrow \mathsf{AbsC}(a *_{\mathrm{C}} b), \\ &\Longrightarrow \mathsf{AbsolutelySummable}(a *_{\mathrm{C}} b). \end{aligned}$$

The public theorem has the textbook premise, while the quantitative estimates still use explicit bounds locally. This is a good example of truncation serving abstraction rather than erasing information prematurely.

5.3 Arbitrary-real scaling from native completeness

The old Cauchy proof scaled a modulus by a rational and therefore exported only

$$q : \mathbb{Q},\ \mathsf{CauchySeries}(a) \Longrightarrow \mathsf{CauchySeries}(qa).$$

In $\mathbb{R}_{\mathrm{S}}$, a constant real sequence is radiuswise Cauchy, multiplication preserves native Cauchy sequences, and the partial sums of ca are eventually equal to c times the partial sums of a. Hence

$$c : \mathbb{R}_{\mathrm{S}},\ \mathsf{CauchySeries}(a) \Longrightarrow \mathsf{CauchySeries}(ca),$$

with the analogous absolute result. This is a genuine extension of the scalar domain, not only a change in witness packaging.

5.4 Uniformity is retained, global choice is not

The location of truncation in the new uniform-series predicate is deliberate:

$$\prod_{\varepsilon} \left\| \sum_{N} \prod_{x} \prod_{m,n \geq N} S_m(x) \sim_{\varepsilon} S_n(x) \right\|.$$

Placing $\prod_x$ *inside* the truncation retains a single cutoff common to the whole family. Placing it outside would yield only pointwise Cauchyness. Placing the outer $\prod_\varepsilon$ inside one truncation would restore a globally chosen modulus. The new definition occupies the exact middle position needed by uniform-limit proofs.

At a requested output radius, the continuity proof opens one common approximation index and one input radius for that single approximant. Both eliminations land in propositions. This is the same radius-by-radius discipline as the completion constructor, propagated through a nontrivial analytic construction.

6 Integration: what is stronger and what is only newer

The $\mathbb{R}_S$ integration layer contains two distinct accomplishments.

1. **Stronger interface discipline.** Riemann resampling and dyadic refinement use $\mathsf{RU}(f)$, not a selected uniform modulus. Vanishing-mesh quadrature sequences are proved radiuswise Cauchy and completed directly. Integrability, uniqueness, linearity, endpoint agreement, and scheme independence are proposition-valued where possible.

2. **New completion-level uniform-limit construction.** For $f_n, f : \mathbb{R}_S \to \mathbb{R}_S$, define uniform convergence by

$$\prod_{\varepsilon:\mathbb{Q}_+} \left\| \sum_{N:\mathbb{N}} \prod_{x:\mathbb{R}_S} \prod_{n \geq N} f_n(x) \sim_\varepsilon f(x) \right\|.$$

   If every f_n is radiuswise uniformly continuous, the limit f is too, and

$$\mathsf{isLim}(n \mapsto I_{a,b}(f_n),\ I_{a,b}(f))$$

   holds. A local theorem assumes convergence only on $[a, b]$; it uses a merely existing rational bound on $b - a$ inside the proposition. A second construction first completes the Cauchy sequence $I_{a,b}(f_n)$ and proves that independent completion agrees with $I_{a,b}(f)$.

These are stronger than the $\mathbb{R}_P$ situation, which has no integration layer, and their hypotheses are constructively sharper than a chosen-modulus formulation. They do not yet subsume the mature $\mathbb{R}_C$ calculus. In particular, no analogue of `HITs/CauchyReals/Calculus/Integration/FundamentalTheorem.agda` was found in $\mathbb{R}_S$.

7 Coverage comparison: do not confuse sharpness with maturity

Scope caveat 1. *The new completion should not be described as simply "stronger than `HITs/CauchyReals`." Its* sequential propositions and theorems that consume them *are more constructively economical. The older $\mathbb{R}_C$ hierarchy nevertheless contains many major developments that have not been transported or reconstructed over $\mathbb{R}_S$.*

Topic	$\mathbb{R}_C$ / $\mathbb{R}_P$ **status**	$\mathbb{R}_S$ **status**
Sequential completeness	$\mathbb{R}_C$ and $\mathbb{R}_P$ complete chosen-modulus sequences.	Radiuswise Cauchy $\simeq$ convergent; exact tail-choice obstruction for the regular comparison.
Series and power series	$\mathbb{R}_P$ has a large generic hierarchy, but rational contractions, selected bounds, and chosen uniform moduli appear in public hypotheses.	Ported and extended with radiuswise predicates, sharp $\lvert x\rvert < 1$ clients, arbitrary-real scaling, bound-free Mertens, and local/uniform differentiation.
Differential calculus	$\mathbb{R}_C$ has derivative, inverse, mean-value, scaling, uniqueness, and power rules; $\mathbb{R}_P$ has an extensive derivative hierarchy.	Generic derivative algebra plus locally uniform power-series differentiation, including $\exp' = \exp$, $\sin' = \cos$, $\cos' = -\sin$, and hyperbolic derivatives.

Topic	$\mathbb{R}_C$ / $\mathbb{R}_P$ status	$\mathbb{R}_S$ status
Integration	older side broader: $\mathbb{R}_C$ has uniform integration, properties, summation, and a fundamental theorem of calculus.	Radiuswise Riemann/dyadic integration, uniqueness, algebraic laws, scheme independence, and uniform-limit preservation; no fundamental theorem located.
Exponential and logarithm	older side broader: $\mathbb{R}_C$ has normalized exponentials, global exp/log laws, logarithm order and derivative theory, rational and real powers.	Power-series exponential with addition, positivity/order reflection/injectivity and derivative theory. The logarithm currently located is the local series $-\log(1-x)$ on the open unit ball, not a global inverse to exp.
Trigonometry	older side broader: $\mathbb{R}_C$ has trigonometric identities, π, arcsine, circle charts/equivalences, angle and winding infrastructure.	Power-series sine/cosine with convergence, addition, parity, derivatives and identities; partial tangent and local arctangent/reciprocal layers. No π, periodicity, or global inverse-trigonometric layer was located.
Hyperbolic functions	No comparably named dedicated hierarchy was located in $\mathbb{R}_C$ or $\mathbb{R}_P$.	new coverage: extensive sinh, cosh, tanh definitions, derivatives, addition/subtraction, shape/order, and fundamental identities.
Complex analysis	older side broader: $\mathbb{R}_C$ has complex algebra, metric completeness, modulus, inverse, polar form, complex differentiation, Cauchy–Riemann, complex exponential/trigonometry, and roots of unity.	No complex-number layer over $\mathbb{R}_S$ was located.

8 Source-to-claim ledger

The following are the decisive checked source locations. Paths are relative to `Cubical/`. To keep the ledger readable, H abbreviates `HITs/CauchyReals`, P abbreviates `HITs/CauchyRealsP`, G abbreviates `Relation/Premetric/Completion`, and S abbreviates `Relation/Premetric/Completion/CauchySequence/Reals`.

Foundations and comparison

- `H/Foundation/Base.agda`: HoTT Definition 11.3.2 constructors `rat`, `lim`, and `eqR` (Agda: $\mathsf{eq}_{\mathbb{R}}$).

- `G/CauchySequence/Base.agda`: equations (3)–(4) and the sequential limit constructor.
- `G/Comparison.agda`: chosen, bundled, and radiuswise witness placements; `PositiveRationalChoice`; `SequentialTailChoice`; conditional comparison equivalences.
- `S/Comparison/Regular.agda`: unconditional `fromRegular` and preservation of rational points/closeness.
- `S/Comparison/Regular/StructuredInverse.agda`: `structuredCanonicalInverse≃sequentialTailChoice`, equation (9).

Sequences and series

- `H/Analysis/Sequence/Base.agda`: `IsCauchySequence'` and `fromCauchySequence'`.
- `P/Analysis/Sequence.agda`: chosen-modulus `cauchySequenceLimit`, selected eventual order.
- `S/Analysis/Sequence/Base.agda`: `chosen→radiuswise`, explicit absence of the converse, `isCauchySequence≃sequenceLimit`, continuous preservation, and truncated eventual equality/asymptotic closeness.
- `S/Analysis/Sequence/Order.agda`: merely eventual $\leq/<$ and closedness under limits.
- `P/Analysis/Series/Base.agda` versus `S/Analysis/Series/Base.agda`: one-way chosen Cauchy $\to$ summable versus `isCauchySeries≃summable`.
- `P/Analysis/Series/Operations.agda` and `P/Analysis/Series/Absolute.agda` versus their S counterparts: rational-only versus arbitrary-real scaling.

Majorants, products, and sharp convergence

- `P/Analysis/Series/PowerSeries/Majorant.agda` versus the S file of the same suffix: removal of `NonnegativeCoefficientsmajorant` by `coefficientMajorantIsNonnegative`.
- `P/Analysis/Series/Convolution/Mertens.agda` and `Convolution/Absolute.agda` versus `S/Analysis/Series/ConvolutionAbsolute.agda`: explicit sum bounds versus merely existing internal bounds; stronger absolute conclusion.
- `P/Analysis/Series/Geometric/Comparison.agda` versus the S file of the same suffix: selected `RationalContractionBound` versus sharp $|x| < 1$.
- `P/Analysis/Series/PowerSeries/Convergence.agda` versus the S file of the same suffix: rational contraction clients versus the `...OfAbs<1` theorem family.
- `P/Analysis/Series/Uniform.agda` versus `S/Analysis/Series/Uniform.agda`: selected uniform modulus versus truncated common cutoffs radius by radius.

Integration and transcendental extensions

- `S/Analysis/Integration/UniformContinuity.agda`: $\mathsf{RU}(f)$ and conventional-uniform $\to$ radiuswise-uniform.
- `S/Analysis/Integration/RiemannIntegral/DyadicUniformLimit/Base.agda` and its `Preservation/Global.agda`, `Preservation/Local.agda`, `Completion/Agreement.agda`: uniform-limit continuity, preservation by the dyadic integral, and agreement of independent completion.
- `H/Calculus/Integration/FundamentalTheorem.agda`, `H/Exponentiation/`, `H/Trigonometry/`, `H/ArcSin/`, `H/Circle/`, and `H/Complex/`: the principal older-coverage caveats.

- `S/Analysis/Series/PowerSeries/Exponential/`, `Trigonometric/`, `Hyperbolic/`, `Arctangent/`, and `Logarithm.agda`: the new $\mathbb{R}_{\mathrm{S}}$ transcendental constructions.

9 Relation to the HoTT Book

The local clone at `/Users/marcin/book` was used for terminology and notation. The most relevant places are:

- **Definition 11.2.10**: Cauchy approximations, the $\mathbb{Q}_+$-indexed presentation used in the HIT constructor.
- **§11.3 and Definition 11.3.2**: the higher inductive-inductive Cauchy reals, including equations corresponding to (1) and the path/extensionality constructor.
- **§11.3.4**: Cauchy completeness of the HIT real.
- **§3.7**: propositional truncation; the notation $\|A\|$ used throughout this report follows the book.
- **§3.8, especially equation (3.8.1)**: the axiom of choice and the movement of truncation across dependent products. Equation (8) is the relevant positive-rational instance.
- **Corollary 3.9.2**: unique choice. It does not solve the tail problem, because tail indices are generally not unique; only limits are unique after separatedness.
- **§6.9**: higher-inductive truncations, providing the conceptual background for eliminating merely existing estimates into propositions.

The HoTT Book's central motivation for the HIT reals is to avoid choosing representatives of a Cauchy sequence in a quotient. The $\mathbb{R}_{\mathrm{S}}$ development pushes the same design discipline one level further into ordinary analysis: it avoids choosing all moduli, contraction radii, sum bounds, uniform cutoffs, or continuity radii whenever the theorem consumes only one such witness at a time and returns a proposition.

10 Bottom line

The strongest justified summary is:

1. The $\mathbb{R}_{\mathrm{S}}$ development accepts the radiuswise predicate $\prod_{\varepsilon} \|\mathsf{Tail}_{\varepsilon}\|$, which is constructively weaker data than the old chosen-modulus predicate.
2. At exactly that strength it proves the full equivalence "Cauchy if and only if convergent." This is the root from which most later interface strengthenings grow.
3. It systematically keeps arbitrary quantitative witnesses under truncation and opens them only in proposition-valued clients. The visible payoffs are sharp $|x| < 1$ hypotheses, bound-free Mertens, arbitrary-real scaling, uniform convergence without a global modulus, and radiuswise integration.
4. The missing reverse comparison is mathematically diagnosed: a structured inverse is equivalent to sequential tail choice, with positive-rational propositional choice as a sufficient assumption.
5. The new line is not yet a replacement for all of `HITs/CauchyReals`. The latter remains materially broader in the fundamental theorem of calculus, global logarithm and real exponentiation, π and inverse trigonometry, circle theory, and complex analysis.

References

[1] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, 2013. Local source consulted at `/Users/marcin/book`, especially Chapters 3, 6, and 11.